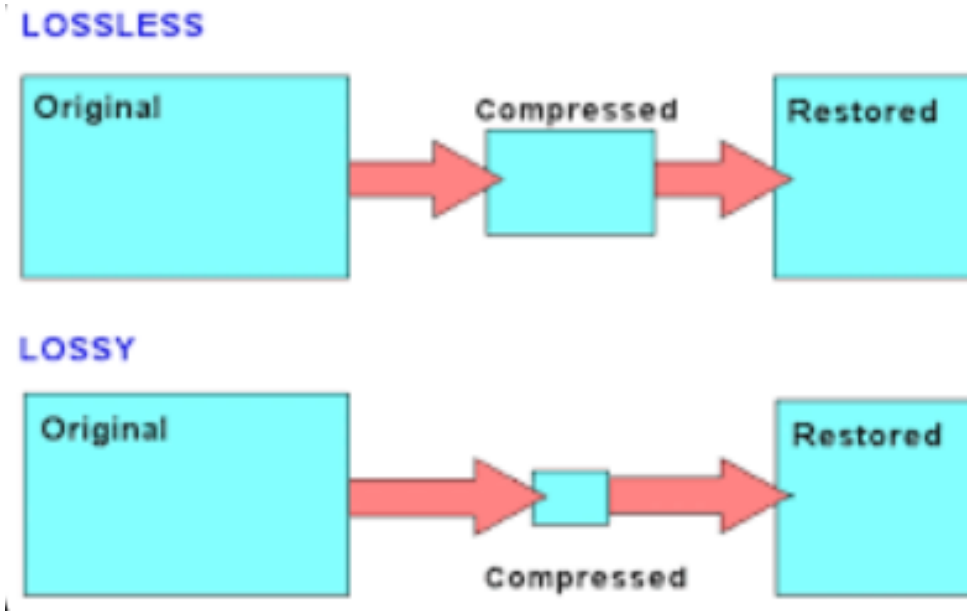
. Lossy Compression Techniques-Predictive Coding and Transform Coding

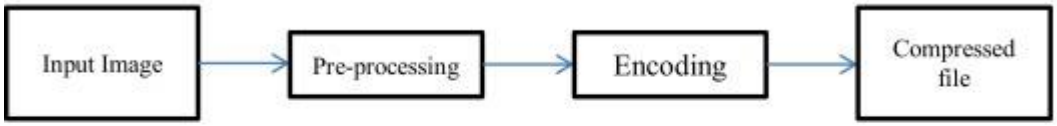
Lossy Compression Techniques

Lossy compression is a fundamental technique in digital media reduces file size by permanently eliminating some data, especially data that is less important for human perception. This is commonly used in **image, audio, and video compression** (e.g., JPEG, MP3, MPEG). Two prominent methods of lossy compression are **Predictive Coding** and **Transform Coding**. While both aim to minimize data, they operate on different principles.

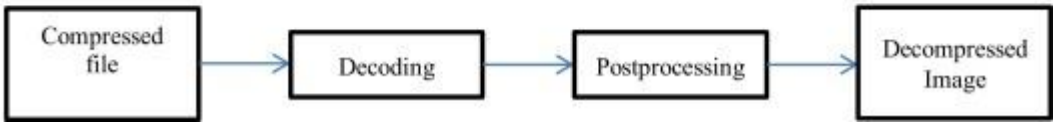
The primary goal of **lossy compression** is to **remove redundant and less noticeable** information from a dataset. This process is **irreversible,** meaning the original data **cannot be perfectly reconstructed** from the compressed version. The effectiveness of lossy compression lies in its ability to achieve **very high compression ratios**, which is crucial for **efficient storage** and transmission of multimedia content.

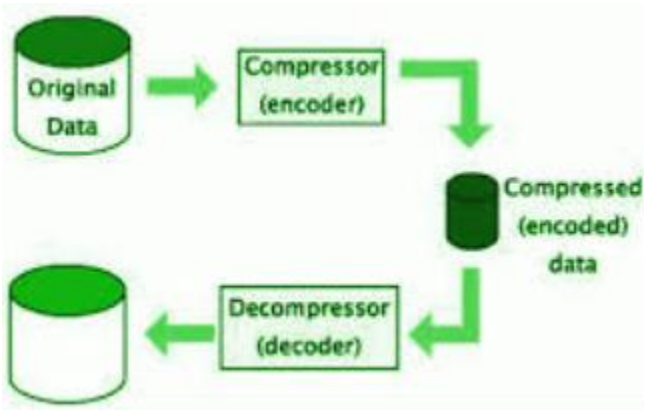


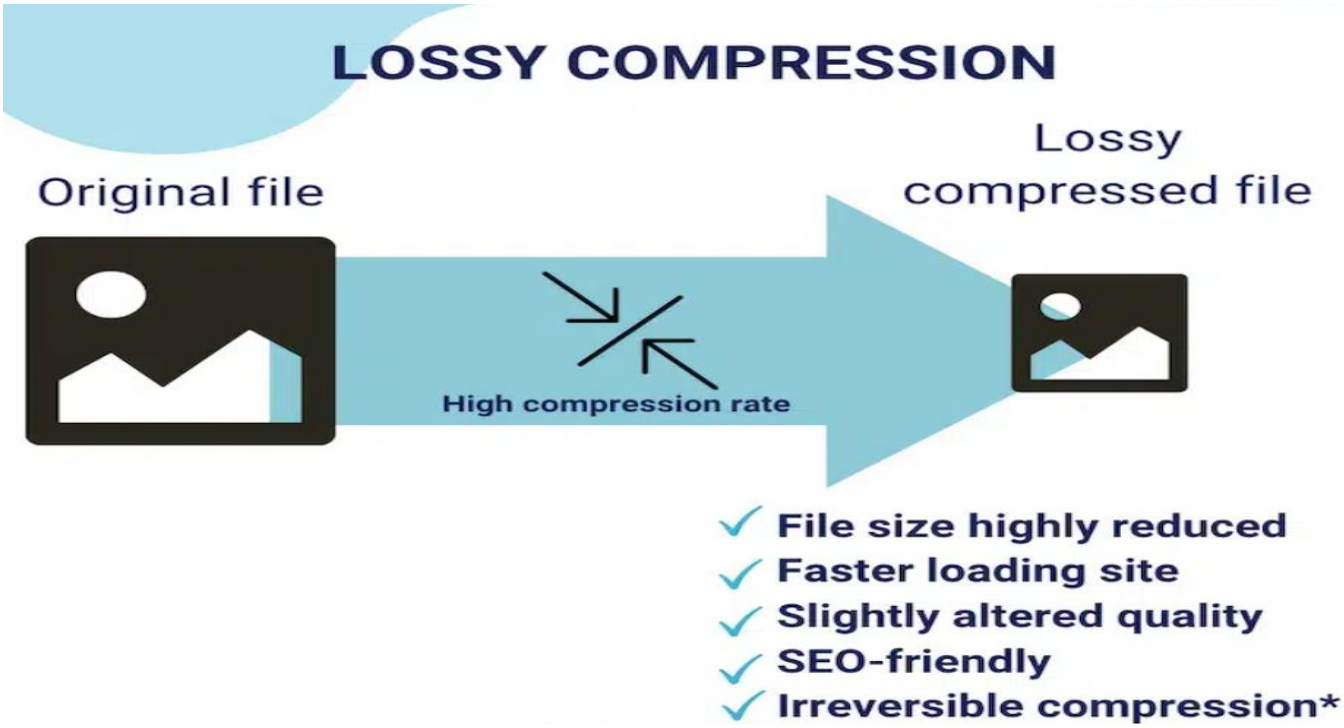
(a)


Input Image
Pre-processing
Encoding
Compressed file

(b)


Compressed file
Decoding
Postprocessing
Decompressed Image


Original Data
Compressor (encoder)
Compressed (encoded) data
Decompressor (decoder)


LOSSY COMPRESSION
Original file
Lossy compressed file
High compression rate
File size highly reduced
Faster loading site
Slightly altered quality
SEO-friendly
Irreversible compression*

The reduced file created by the compression process is called the *compressed file* and is used to reconstruct the image, resulting in the *decompressed image*. The ratio of the original (uncompressed file) and the compressed file is referred to as the *compression ratio*. The compression ratio is determined according to the following equation:

$$Compression\ Ratio = \frac{Uncompressed\ \ file\ \ size\ (bytes)}{Compressed\ \ file\ \ size\ (bytes)}$$

1. Predictive Coding

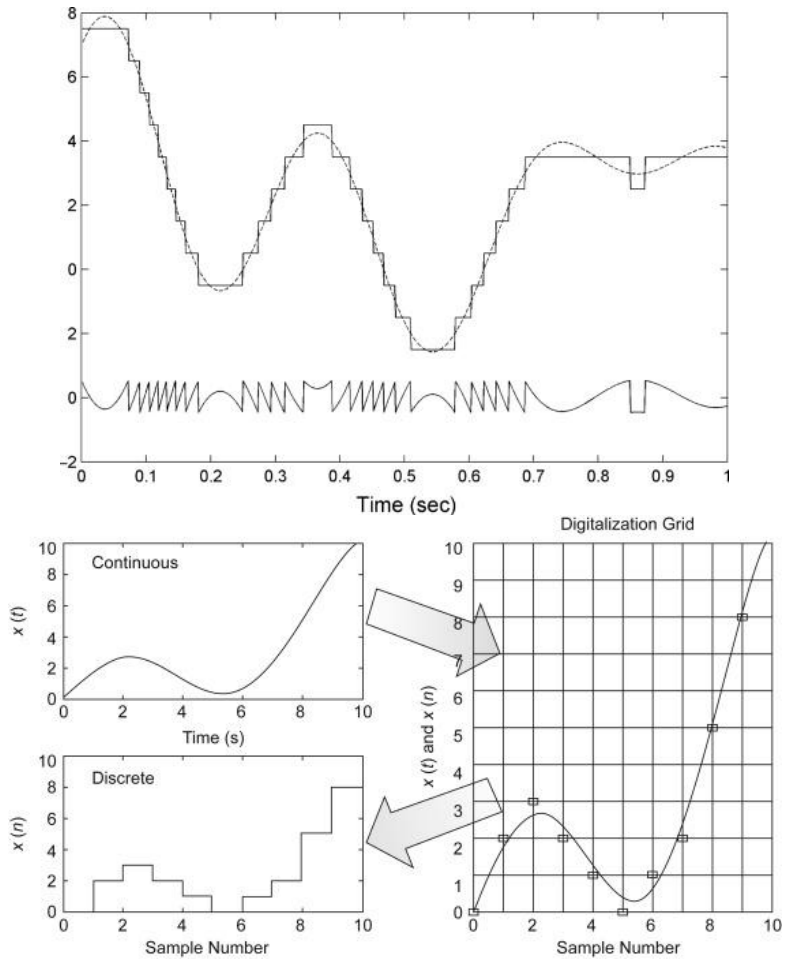
Predictive coding works on the principle of predicting the value of a data point (like a pixel or an audio sample) based on its neighboring or preceding values. Instead of encoding the actual value, the system encodes the **difference** between the predicted value and the actual value, known as the **prediction error** or **residual.**

Since adjacent data points in images and audio are often very similar, the prediction error is typically small and requires fewer bits to represent than the original data. This is particularly effective for reducing what is known as **temporal redundancy** in video frames or **spatial redundancy** in images.

Spatial redundancy, in the context of signal processing and data compression, refers to the correlation or similarity between neighboring elements within a single data structure, like an image or video frame

How Predictive Coding Works:

1. **Prediction:** A predictor model estimates the value of the current sample based on previously encoded samples.
2. **Error Calculation:** The predicted value is subtracted from the actual value to get the prediction error.
3. **Quantization:** The prediction error is quantized. This is where information is lost. A larger quantization step results in greater compression but also more potential for error in the reconstructed signal.
4. **Encoding:** The quantized error is then encoded, often using a variable-length coding scheme where smaller error values get shorter codes.



Types of Predictors:

- **Linear Prediction:** Uses a linear combination of previous samples.
- **Delta Modulation:** Simplest form – predicts the next value is equal to the current one.

Applications:

- Speech compression (e.g., Linear Predictive Coding - LPC)
- Image compression (JPEG-LS)

Advantages:

- Simple to implement.
- Efficient when the signal has strong correlation.

Limitations:

- Performance drops when correlation is low.
- Prediction errors may accumulate.

2. Transform Coding

Concept:

Transform coding converts the original data (typically spatial or time domain) into a different domain (usually frequency domain) where it becomes easier to identify and remove **perceptually insignificant components**. The most widely used transform for this purpose is the **Discrete Cosine Transform (DCT)**, which is the backbone of the JPEG image compression standard and is also used in many video compression standards like MPEG and AVC.

MPEG (Moving Picture Experts Group) is a standard for compressing and encoding digital video and audio files. MPEG files can have extensions like .mpg or .mpeg, but other formats like MP4 also utilize MPEG compression

It's a widely used format for recording, compressing, and distributing digital video. AVC is a key component in technologies like Blu-ray discs and is commonly used by streaming services like YouTube

How Transform Coding Works:

1. **Blocking:** The input data (e.g., an image) is divided into small blocks (e.g., 8x8 pixels).
2. **Transformation:** A mathematical transform, like the DCT, is applied to each block. This converts the spatial pixel values into frequency coefficients. The most significant information is typically concentrated in the top-left corner of the coefficient block (representing low frequencies).
3. **Quantization:** The transformed coefficients are then quantized. High-frequency coefficients, to which the human eye is less sensitive, are quantized more aggressively (i.e., more information is discarded). This is the primary source of data loss in transform coding.
4. **Encoding:** The quantized coefficients are reordered (often in a zigzag pattern to group the zeroed-out high-frequency coefficients) and then losslessly encoded.

Applications:

- Image compression (JPEG)
- Video compression (H.264, MPEG)
- Audio compression (MP3, AAC)

Advantages:

- High compression ratio.
- Good perceptual quality.

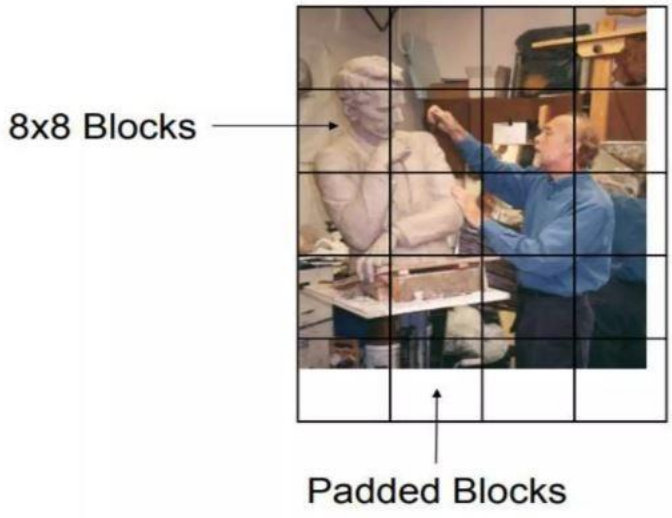
Comparison:

Feature	Predictive Coding	Transform Coding
Domain	Time/Spatial	Frequency/Transform
Compression Method	Encode prediction error	Encode transformed coefficients
Complexity	Low to moderate	High
Best suited for	Audio, speech, simple images	Images, video, multimedia

WHAT IS JPEG STANDARDS???

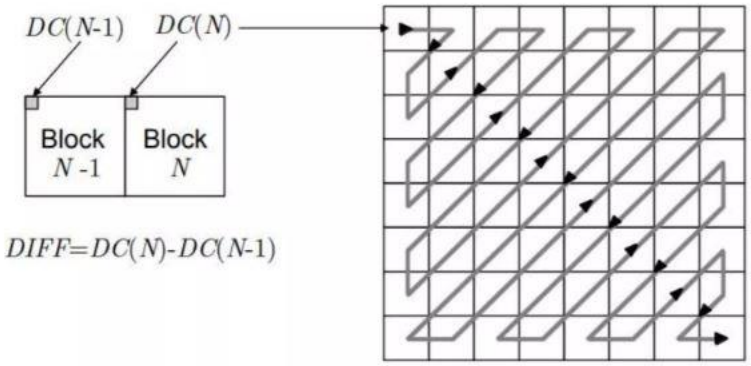
- JPEG is sophisticated lossless compression method for color or grey-scale still images.
- It works best on continuous tone images where adjacent pixels have similar colors.
- One advantage of JPEG is the use of many parameters, allowing the use to adjust the amount of data lost over a wide variety range.

JPEG: Image Partitioning



Differential coding of DC coefficient: DPCM using previous quantized DC coefficient as predictor

Zig-zag scan of AC coefficients



ORIGINAL vs LOSSLESS vs LOSSY
Same visual quality
ORIGINAL
650 KB
LOSSLESS
625 KB
LOSSY
100 KB

Feature	Lossy Compression	Lossless Compression
File Size Reduction	Smaller file size	Larger file size
Data Integrity	Some data is removed but usually (invisible to the naked eye)	Full data integrity
Positive Impact on Page Speed	High and more bandwidth saved	Low and less bandwidth saved
Compression Efficiency	High	Low
Reversibility	Irreversible (cannot restore the original file)	Reversible (original can be restored)
Visual Quality	Minor, usually unnoticeable changes	No change at all
Best Use Cases	Web images, WooCommerce product images, Photo galleries and media files	Photography, documents, software, backups